you've already admitted that the wave function is part of reality. The particle positions, for that matter, have absolutely no influence on how the wave function evolves. All they do, arguably, is point to a particular branch of the wave function and say, "This is the real one." Some Everettians have therefore claimed that Bohmian mechanics isn't really any different from Everett, it just includes some superfluous extra variables that serve no purpose but to assuage some anxieties about splitting into multiple copies of ourselves. As Deutsch has put it, "Pilot-wave theories are parallel-universe theories in a state of chronic denial."

We won't adjudicate this dispute right here. What's clear is that Bohmian mechanics is an explicit construction that does what many physicists thought was impossible: to construct a precise, deterministic theory that reproduces all of the predictions of textbook quantum mechanics, without requiring any mysterious incantations about the measurement process or a distinction between quantum and classical realms. The price we pay is explicit nonlocality in the dynamics.

Bohm was hopeful that his new theory would be widely appreciated by physicists. This was not to be. In the emotionally charged language that so often accompanies discussions of quantum foundations, Heisenberg called Bohm's theory "a superfluous ideological superstructure," while Pauli referred to it as "artificial metaphysics." We've already heard the judgment of Oppenheimer, who had previously been Bohm's mentor and supporter. Einstein seems to have appreciated Bohm's effort, but thought the final construction was artificial and unconvincing. Unlike de Broglie, however, Bohm didn't bow to the pressure, and continued to develop and advocate for his theory. Indeed, his advocacy inspired de Broglie himself, who was still around and active (he died in 1987). In his later years de Broglie returned to hidden-variable theories, developing and elaborating his original model.

Even apart from the presence of explicit nonlocality and the accusation that the theory is just Many-Worlds in denial, there are other significant problems inherent in Bohmian mechanics, especially from the perspective of a modern fundamental physicist. The list of ingredients in the theory is